

Total No. of Printed Pages:2

SUBJECT CODE NO: - RNP-8009

FACULTY OF SCIENCE AND TECHNOLOGY

F.E (Group A & B) (NEP) Sem - I and Sem - II

Examination November / December 2025 (To be held in February-2026)

Engineering Chemistry

[Time: 3:00 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no. 1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve the following questions (any 10)

20

- a) CNTs are mainly used for their excellent
 - A) Color
 - B) Electrical and mechanical properties
 - C) Toxicity
 - D) Odor
- b) Bakelite is a _____ type of polymer
- c) Which of the following is a reinforcement material?
 - A) Bakelite
 - B) Silicon carbide
 - C) PVC
 - D) PVA
- d) Give synthesis of PVA
- e) Define cloud point and pour point of lubricating oil
- f) Define coolant give its two examples
- g) Differentiate between dry corrosion and wet corrosion
- h) What is electroplating give its applications
- i) Roasting is generally done for
 - A) Sulphide ores
 - B) Metal alloys
 - C) Polymers
 - D) Carbon materials
- j) Formula to calculate Viscosity index is.....
- k) Explain any two advantages of lithium-ion batteries
- l) Draw diagram of fuel cell

SECTION B

Q2 A. Explain Caustic embrittlement

05

B. Give synthesis properties and applications of PVC

05

- Q3 A. Write a note on ion exchange processes 05
 B. Explain properties and uses of Ethylene glycol as coolant 05
- Q4 A. Writes a note on wet corrosion 05
 B. Explain Computed Radiography and digital Radiography 05
- Q5 A. Explain carbon -carbon, ceramic matrix composites 05
 B. Calculate total hardness of water containing following impurities 05
- | Impurity | Wt | Molecular wt. |
|---------------|---------|---------------|
| CO_2 | 13.5ppm | 44 |
| $CaSO_4$ | 65.2ppm | 136 |
| $Ca(HCO_3)_2$ | 21ppm | 162 |
| $MgCl_2$ | 10ppm | 95 |
| KCl | 5ppm | 74.5 |
- Q6 A. Explain composition, properties, and applications of LNG 05
 B. Kerosene contain 12% hydrogen and its GCV IS 45 mJ/Kg find net calorific value of kerosene 05
- Q7 A. How corrosion is protected by 05
 a) Sacrificial anodic protection b) Electroplating
 B. Define a) Laser cladding b) Casting c) Rolling d) Machining e) Sintering 05

